MOS Capacitor Simulation project

Corrected equations (as implemented in the code)

1 Model and notation

The structure is gate – oxide – silicon (p-type or n-type) – grounded bulk contact, in one dimension x.

$$V_t = \frac{kT}{q}, \qquad \phi = \frac{V}{V_t}, \qquad E = -\frac{dV}{dx} \tag{1}$$

The grid nodes are $x_0 \dots x_N$. Node 0 is the gate, node s is the oxide–semiconductor interface and node N is the bulk contact. The oxide mesh is uniform; in the semiconductor the spacing grows geometrically:

$$\Delta x_i = x_{i+1} - x_i, \qquad \Delta x_{s+k} = a\, b^k \quad \Longrightarrow \quad x_{s+k} = t_{ox} + a\,\frac{b^k - 1}{b - 1} \tag{2}$$

Each node owns a control volume. Its semiconductor part has width

$$h_i = \begin{cases} 0 & i < s \quad \text{(oxide)} \\ \tfrac{1}{2}\Delta x_s & i = s \quad \text{(interface: half a cell)} \\ \tfrac{1}{2}\left(\Delta x_{i-1} + \Delta x_i\right) & s < i < N \end{cases} \tag{3}$$

Differences of the normalised potential between neighbouring nodes are written $\Delta\phi_i = \phi_{i+1} - \phi_i$.

2 Current densities (Scharfetter–Gummel)

$$J_n = q\mu_n n E + qD_n\frac{dn}{dx} = kT\mu_n\left(-n\frac{d\phi}{dx} + \frac{dn}{dx}\right) \tag{4}$$

$$J_p = q\mu_p p E - qD_p\frac{dp}{dx} = -kT\mu_p\left(p\frac{d\phi}{dx} + \frac{dp}{dx}\right) \tag{5}$$

The current is a flux between two nodes, so it is defined on the element $i+\frac{1}{2}$ between nodes i and $i{+}1$. Taking the potential linear inside an element and writing $n = u\,e^{\phi}$ gives $J_n = kT\mu_n e^{\phi}\,du/dx$. Integrating $J_n e^{-\phi}$ over the element with constant J_n:

$$J_{n,i+\frac{1}{2}} = \frac{kT\mu_n}{\Delta x_i}\,B(\Delta\phi_i)\left(n_{i+1} - n_i e^{\Delta\phi_i}\right) = \frac{kT\mu_n}{\Delta x_i}\left[B(\Delta\phi_i)\,n_{i+1} - B(-\Delta\phi_i)\,n_i\right] \tag{6}$$

with the Bernoulli function

$$B(x) = \frac{x}{e^x - 1}, \qquad B(x)\,e^x = B(-x) \tag{7}$$

For holes take $p = u\,e^{-\phi}$ (note the minus sign), so $J_p = -kT\mu_p e^{-\phi}\,du/dx$ and

$$J_{p,i+\frac{1}{2}} = \frac{kT\mu_p}{\Delta x_i}\,B(\Delta\phi_i)\left(p_i - p_{i+1}e^{\Delta\phi_i}\right) = \frac{kT\mu_p}{\Delta x_i}\left[B(\Delta\phi_i)\,p_i - B(-\Delta\phi_i)\,p_{i+1}\right] \tag{8}$$

3 Steady state (equilibrium and DC bias)

3.1 Poisson's equation

$$\frac{d}{dx}\left(\varepsilon\frac{dV}{dx}\right) = -\rho, \qquad \rho = q\,(N_D - N_A + p - n) \tag{9}$$

Integrating over the control volume of node i (there is no charge in the oxide, where $h_i = 0$):

$$F_i \equiv \varepsilon_{i+\frac{1}{2}}\frac{V_{i+1} - V_i}{\Delta x_i} - \varepsilon_{i-\frac{1}{2}}\frac{V_i - V_{i-1}}{\Delta x_{i-1}} + q\,h_i\,(N_D - N_A + p_i - n_i) + Q_f\,\delta_{i,s} = 0 \tag{10}$$

where $\varepsilon_{i+\frac{1}{2}} = \varepsilon_{ox}$ for $i < s$ and ε_{si} for $i \geq s$. At the interface node $i = s$ the two permittivities differ, which enforces continuity of εE; the fixed oxide charge Q_f (per unit area) sits at this node only.

Boundary conditions. V is measured from the bulk, so $V_N = 0$. The gate and the semiconductor have different work functions, so a gate voltage V_G puts the gate node at

$$V_0 = V_G - \phi_{ms}, \qquad \phi_{ms} = \phi_m - \phi_s, \qquad \phi_s = \chi + \frac{E_g}{2q} - V_t \ln\frac{n_{bulk}}{n_{int}} \tag{11}$$

with ϕ_m the gate work function, χ the electron affinity and n_{bulk} the electron concentration at the bulk contact. This covers both substrate types: $N_A > N_D$ is p-type ($n_{bulk} < n_{int}$) and $N_D > N_A$ is n-type. The bands are flat ($V = 0$ everywhere) at the flat-band voltage

$$V_{FB} = \phi_{ms} - \frac{Q_f}{C_{ox}}, \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \tag{12}$$

Equilibrium is $V_G = 0$, which is no longer the flat-band condition.

The equation is non-linear because n and p depend exponentially on V. One Newton step $J\,\Delta V = -F$ is taken with the carriers following the potential at fixed quasi-Fermi levels, $n \propto e^{V/V_t}$, $p \propto e^{-V/V_t}$:

$$J_{i,i\pm1} = \frac{\varepsilon_{i\pm\frac{1}{2}}}{\Delta x_i \text{ or } \Delta x_{i-1}}, \qquad J_{i,i} = -\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x_i} - \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x_{i-1}} - \frac{q\,h_i\,(n_i + p_i)}{V_t} \tag{13}$$

followed by $V \leftarrow V + \Delta V$, $n \leftarrow n\,e^{\Delta V/V_t}$, $p \leftarrow p\,e^{-\Delta V/V_t}$. Steps larger than V_t are damped to $\pm V_t \ln(1 + |\Delta V|/V_t)$.

3.2 Carrier continuity

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} - R, \qquad \frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} - R \tag{14}$$

Recombination is Shockley–Read–Hall, with n_{int} the intrinsic carrier concentration:

$$R = \frac{np - n_{int}^2}{S}, \qquad S = \tau_p\,(n + n_{int}) + \tau_n\,(p + n_{int}) \tag{15}$$

(A relaxation term $(n - n_{eq})/\tau_n$ with a fixed n_{eq} must not be used here: under bias it pulls the carriers back to their zero-bias values, although the true steady state of a MOS capacitor has $R = 0$ with $np = n_{int}^2$.) In steady state, integrating over the control volume of node i:

$$\frac{1}{q}\left(J_{n,i+\frac{1}{2}} - J_{n,i-\frac{1}{2}}\right) = h_i R_i, \qquad -\frac{1}{q}\left(J_{p,i+\frac{1}{2}} - J_{p,i-\frac{1}{2}}\right) = h_i R_i \tag{16}$$

Inserting (6), (8) and dividing by $V_t\mu$, with S_i and the other carrier taken from the previous iteration:

$$\alpha_{i-1}n_{i-1} + \alpha_i n_i + \alpha_{i+1}n_{i+1} = -\frac{h_i\, n_{int}^2}{V_t\mu_n S_i} \tag{17}$$

$$\alpha_{i+1} = \frac{B(\Delta\phi_i)}{\Delta x_i}, \quad \alpha_{i-1} = \frac{B(-\Delta\phi_{i-1})}{\Delta x_{i-1}}, \quad \alpha_i = -\left[\frac{B(-\Delta\phi_i)}{\Delta x_i} + \frac{B(\Delta\phi_{i-1})}{\Delta x_{i-1}}\right] - \frac{h_i\, p_i}{V_t\mu_n S_i} \tag{18}$$

$$\beta_{i-1}p_{i-1} + \beta_i p_i + \beta_{i+1}p_{i+1} = -\frac{h_i\, n_{int}^2}{V_t\mu_p S_i} \tag{19}$$

$$\beta_{i+1} = \frac{B(-\Delta\phi_i)}{\Delta x_i}, \quad \beta_{i-1} = \frac{B(\Delta\phi_{i-1})}{\Delta x_{i-1}}, \quad \beta_i = -\left[\frac{B(\Delta\phi_i)}{\Delta x_i} + \frac{B(-\Delta\phi_{i-1})}{\Delta x_{i-1}}\right] - \frac{h_i\, n_i}{V_t\mu_p S_i} \tag{20}$$

The recombination term enters α_i, β_i and the right-hand side with a *minus* sign.

Boundary conditions:

- Interface ($i = s$): no current enters the oxide, $J_{s-\frac{1}{2}} = 0$. The equations above are used with the $i-1$ terms dropped and $h_s = \frac{1}{2}\Delta x_s$. For $\Delta x_s \to 0$ this is $n_{s+1} = n_s e^{\Delta\phi_s}$ and $p_{s+1}e^{\Delta\phi_s} = p_s$.
- Bulk contact ($i = N$): n_N, p_N keep the charge-neutral equilibrium values, from $p - n = N_A - N_D$ and $np = n_{int}^2$.

Poisson's equation and the two continuity equations are solved one after the other and repeated until V, n and p stop changing. The gate voltage is moved in steps of at most 0.1 V, each step starting from the previous solution.

3.3 Solving the continuity equations accurately

Both equations have the form $\mathcal{F}_i - \mathcal{F}_{i-1} - \sigma_i y_i = -g_i$ with flux $\mathcal{F}_i = u_i y_{i+1} - d_i y_i$ and $u_i, d_i, \sigma_i, g_i \geq 0$. The fluxes are up to 10^{17} times larger than the recombination terms, so ordinary Gaussian elimination loses them to round-off. The elimination is therefore arranged so that it never subtracts: after eliminating nodes $s \ldots i$, node i obeys $u_i y_{i+1} - (d_i + e_i)\, y_i = -b_i$ with

$$e_s = \sigma_s, \quad b_s = g_s, \qquad e_{i+1} = \sigma_{i+1} + \frac{u_i\, e_i}{d_i + e_i}, \qquad b_{i+1} = g_{i+1} + \frac{d_i\, b_i}{d_i + e_i} \tag{21}$$

and the back substitution is $y_i = (u_i y_{i+1} + b_i) \,/\, (d_i + e_i)$.

4 Small-signal AC analysis

A small sinusoidal signal is added to the DC gate bias, and every quantity is written as its DC value plus a phasor:

$$V_G = V_{DC} + \tilde{v}\, e^{j\omega t}, \qquad V = V_0 + \tilde{V} e^{j\omega t}, \qquad n = n_0 + \tilde{n}\, e^{j\omega t}, \qquad p = p_0 + \tilde{p}\, e^{j\omega t} \tag{22}$$

with $\Delta\tilde{\phi}_i = (\tilde{V}_{i+1} - \tilde{V}_i)/V_t$. Only terms linear in the perturbation are kept, using

$$B(\Delta\phi_i + \Delta\tilde{\phi}_i e^{j\omega t}) \simeq B(\Delta\phi_{0,i}) + B'(\Delta\phi_{0,i})\, \Delta\tilde{\phi}_i e^{j\omega t}, \qquad B'(x) = \frac{B(x)\,(1 - B(-x))}{x} \tag{23}$$

4.1 Linearised currents

From (6), $J_{n,i+\frac{1}{2}} = J_{0,i+\frac{1}{2}} + \tilde{J}_{n,i+\frac{1}{2}} e^{j\omega t}$ with

$$\tilde{J}_{n,i+\frac{1}{2}} = \frac{kT\mu_n}{\Delta x_i}\Big[B(\Delta\phi_{0,i})\,\tilde{n}_{i+1} - B(-\Delta\phi_{0,i})\,\tilde{n}_i + g_{n,i}\,\Delta\tilde{\phi}_i\Big] \tag{24}$$

$$g_{n,i} = B'(\Delta\phi_{0,i})\,n_{0,i+1} + B'(-\Delta\phi_{0,i})\,n_{0,i} \tag{25}$$

This is the same as $B(\Delta\phi_{0,i})\big[\tilde{n}_{i+1} - e^{\Delta\phi_{0,i}}(\tilde{n}_i + n_{0,i}\Delta\tilde{\phi}_i)\big] + B'(\Delta\phi_{0,i})\,\Delta\tilde{\phi}_i\big[n_{0,i+1} - n_{0,i}e^{\Delta\phi_{0,i}}\big]$, because $-\frac{d}{dx}\big[B(x)e^x\big] = B'(-x)$. From (8):

$$\tilde{J}_{p,i+\frac{1}{2}} = \frac{kT\mu_p}{\Delta x_i}\Big[B(\Delta\phi_{0,i})\,\tilde{p}_i - B(-\Delta\phi_{0,i})\,\tilde{p}_{i+1} + g_{p,i}\,\Delta\tilde{\phi}_i\Big] \tag{26}$$

$$g_{p,i} = B'(\Delta\phi_{0,i})\,p_{0,i} + B'(-\Delta\phi_{0,i})\,p_{0,i+1} \tag{27}$$

4.2 Linearised continuity equations

Linearising (15) around the DC solution (R_0, S_0 are the DC values):

$$\tilde{R} = r_n\tilde{n} + r_p\tilde{p}, \qquad r_n = \frac{p_0 - R_0\tau_p}{S_0}, \qquad r_p = \frac{n_0 - R_0\tau_n}{S_0} \tag{28}$$

The DC parts cancel, and $\partial/\partial t \to j\omega$. For node i:

$$\frac{1}{q}\left(\tilde{J}_{n,i+\frac{1}{2}} - \tilde{J}_{n,i-\frac{1}{2}}\right) - h_i\,(r_{n,i} + j\omega)\,\tilde{n}_i - h_i r_{p,i}\,\tilde{p}_i = 0 \tag{29}$$

$$-\frac{1}{q}\left(\tilde{J}_{p,i+\frac{1}{2}} - \tilde{J}_{p,i-\frac{1}{2}}\right) - h_i\,(r_{p,i} + j\omega)\,\tilde{p}_i - h_i r_{n,i}\,\tilde{n}_i = 0 \tag{30}$$

Written out with (24), and $\alpha_{i\pm1}$ as in the DC case evaluated at the DC potential:

$$\alpha_{i+1}\tilde{n}_{i+1} + \left[\alpha_i^{flux} - \frac{h_i\,(r_{n,i} + j\omega)}{V_t\mu_n}\right]\tilde{n}_i + \alpha_{i-1}\tilde{n}_{i-1} - \frac{h_i\,r_{p,i}}{V_t\mu_n}\tilde{p}_i + c_{n,i} = 0 \tag{31}$$

$$\alpha_i^{flux} = -\left[\frac{B(-\Delta\phi_{0,i})}{\Delta x_i} + \frac{B(\Delta\phi_{0,i-1})}{\Delta x_{i-1}}\right], \qquad c_{n,i} = \frac{g_{n,i}\,\Delta\tilde{\phi}_i}{\Delta x_i} - \frac{g_{n,i-1}\,\Delta\tilde{\phi}_{i-1}}{\Delta x_{i-1}} \tag{32}$$

and the same for holes with (26):

$$\beta_{i+1}\tilde{p}_{i+1} + \left[\beta_i^{flux} - \frac{h_i\,(r_{p,i} + j\omega)}{V_t\mu_p}\right]\tilde{p}_i + \beta_{i-1}\tilde{p}_{i-1} - \frac{h_i\,r_{n,i}}{V_t\mu_p}\tilde{n}_i + c_{p,i} = 0 \tag{33}$$

$$\beta_i^{flux} = -\left[\frac{B(\Delta\phi_{0,i})}{\Delta x_i} + \frac{B(-\Delta\phi_{0,i-1})}{\Delta x_{i-1}}\right], \qquad c_{p,i} = -\frac{g_{p,i}\,\Delta\tilde{\phi}_i}{\Delta x_i} + \frac{g_{p,i-1}\,\Delta\tilde{\phi}_{i-1}}{\Delta x_{i-1}} \tag{34}$$

4.3 Linearised Poisson equation

c_n, c_p contain $\tilde{V}$, and $\tilde{V}$ depends on $\tilde{n}$, $\tilde{p}$ through the charge, so Poisson's equation (10) is linearised and solved together with (29), (30):

$$\varepsilon_{i+\frac{1}{2}}\frac{\tilde{V}_{i+1} - \tilde{V}_i}{\Delta x_i} - \varepsilon_{i-\frac{1}{2}}\frac{\tilde{V}_i - \tilde{V}_{i-1}}{\Delta x_{i-1}} + q\,h_i\,(\tilde{p}_i - \tilde{n}_i) = 0 \tag{35}$$

4.4 Boundary conditions

- Gate: $\tilde{V}_0 = \tilde{v}$. The oxide has no charge, so $\tilde{V}$ is linear in it and the oxide acts as the capacitor $C_{ox} = \varepsilon_{ox}/t_{ox}$ between gate and interface.
- Interface ($i = s$): $\tilde{J}_{n,s-\frac{1}{2}} = \tilde{J}_{p,s-\frac{1}{2}} = 0$; equations (29), (30) are used with $h_s = \frac{1}{2}\Delta x_s$. For $\Delta x_s \to 0$ this is $\tilde{J}_{s+\frac{1}{2}} = 0$.
- Bulk contact ($i = N$): $\tilde{V}_N = 0$, $\tilde{n}_N = \tilde{p}_N = 0$.

4.5 Solution

The AC problem is linear, so it is solved once per bias point and frequency, with no iteration. ϕ_{ms} and Q_f are constants, so they do not appear in the AC equations.

Solving directly for $\tilde{n}$, $\tilde{p}$ is badly conditioned in strong inversion, because the drift term $g\,\Delta\tilde{\phi}$ and the diffusion terms of (24) nearly cancel. The code therefore uses the AC quasi-Fermi potentials $\tilde{W}_n$, $\tilde{W}_p$ as unknowns:

$$\tilde{n} = \frac{n_0}{V_t}\left(\tilde{V} - \tilde{W}_n\right), \qquad \tilde{p} = \frac{p_0}{V_t}\left(\tilde{W}_p - \tilde{V}\right) \tag{36}$$

No DC current flows in a MOS capacitor, so $B(\Delta\phi_{0,i})\,n_{0,i+1} = B(-\Delta\phi_{0,i})\,n_{0,i} \equiv \bar{n}_i$, and then $g_{n,i} = -\bar{n}_i$. Likewise $B(\Delta\phi_{0,i})\,p_{0,i} = B(-\Delta\phi_{0,i})\,p_{0,i+1} \equiv \bar{p}_i = -g_{p,i}$. Inserting these in (24), (26), the cancellation is done analytically and the currents become plain conductances:

$$\frac{\tilde{J}_{n,i+\frac{1}{2}}}{q} = -\frac{\mu_n \bar{n}_i}{\Delta x_i}\left(\tilde{W}_{n,i+1} - \tilde{W}_{n,i}\right), \qquad \frac{\tilde{J}_{p,i+\frac{1}{2}}}{q} = -\frac{\mu_p \bar{p}_i}{\Delta x_i}\left(\tilde{W}_{p,i+1} - \tilde{W}_{p,i}\right) \tag{37}$$

These are the same equations as (29), (30), (35), only written in other variables. With $X_i = (\tilde{V}_i, \tilde{W}_{n,i}, \tilde{W}_{p,i})^T$, the three equations of node i are

$$(U_i X_{i+1} - D_i X_i) - (U_{i-1} X_i - D_{i-1} X_{i-1}) - L_i X_i = 0 \tag{38}$$

with 3×3 complex matrices: $U_i = D_i = \mathrm{diag}(\varepsilon_{si},\ \mu_n\bar{n}_i,\ \mu_p\bar{p}_i)\,/\Delta x_i$ hold the flux coefficients of element $i + \frac{1}{2}$ and L_i the local terms (charge, recombination, $j\omega$). As in the DC case the fluxes are never subtracted from each other. After eliminating nodes $s \ldots i$, node i obeys $U_i X_{i+1} - (D_i + E_i) X_i = -b_i$ with

$$E_s = L_s + \mathrm{diag}(C_{ox}, 0, 0), \qquad b_s = (C_{ox}\tilde{v}, 0, 0)^T \tag{39}$$

$$E_{i+1} = L_{i+1} + E_i\,(D_i + E_i)^{-1} U_i, \qquad b_{i+1} = D_i\,(D_i + E_i)^{-1} b_i \tag{40}$$

and back substitution $X_i = (D_i + E_i)^{-1}\,(U_i X_{i+1} + b_i)$ starting from $X_N = 0$.

Two kinds of sweep are done with this solver: a bias sweep (C and G/ω versus V_G at a few frequencies) and a frequency sweep (C and G/ω versus f at a few gate voltages).

5 Capacitance

The gate charge per unit area is $Q = \varepsilon_{ox} E_{ox}$, so its phasor is $\tilde{Q} = \varepsilon_{ox}(\tilde{V}_0 - \tilde{V}_1)/\Delta x_0 = C_{ox}(\tilde{v} - \tilde{V}_s)$ and

$$\tilde{C} = \frac{\tilde{Q}}{\tilde{v}} = C_{ox}\left(1 - \frac{\tilde{V}_s}{\tilde{v}}\right), \qquad C = \mathrm{Re}\,\tilde{C}, \qquad G = -\omega\,\mathrm{Im}\,\tilde{C} \tag{41}$$

C is the capacitance and G the conductance per unit area at angular frequency ω. At low frequency the inversion layer follows the signal and C returns to about C_{ox} in inversion; at high frequency it cannot, and C stays at its minimum.

6 Corrections made to `eq.pdf`

1. DC recombination term: the sign was wrong in α_i, β_i and the right-hand sides (old eqs. 16–23). It now enters with a minus sign, consistent with the AC equations.

2. Recombination model: $(n - n_{eq})/\tau$ replaced by SRH, (15), in both DC and AC.

3. c_n (below old eq. 40): two sign errors. The $B\,n_{0,i} e^{\Delta\phi_{0,i}} \Delta\tilde{\phi}_i/\Delta x_i$ term is negative and the B' term of element $i - \frac{1}{2}$ is negative.

4. c_p: the last bracket was $(p_{0,i} - p_{0,i}e^{\Delta\phi_{0,i}})$; it is $(p_{0,i} - p_{0,i+1}e^{\Delta\phi_{0,i}})$.

5. Holes: "take $p = u\,e^{\phi}$" is $p = u\,e^{-\phi}$.

6. Control volume: $\Delta x_{i-1/2}$ is now defined, as $h_i = \frac{1}{2}(\Delta x_{i-1} + \Delta x_i)$.

7. Interface boundary: the half control volume of the interface node is kept, instead of setting $\tilde{J} = 0$ one element inside the semiconductor.

8. Added: discretised Poisson equation (DC and linearised AC), the coupling of the three AC equations, the capacitance formula, and the missing factor q in ρ.